Data Flow Diagram

Date	21 October 2026
Team ID	VAI-AI-2026
Project Name	VoyageAI — Personalized AI Travel Itinerary Planner
Maximum Marks	2 Marks

Data Flow Diagram (DFD) Specification:

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

Symbol	Name	Description
Oval / Rounded	External Entity	A person, system, or organization outside the project boundary (e.g., Traveler, Groq Cloud AI API).
Rectangle (Numbered)	Process	An activity transforming incoming data to outgoing data (e.g., Validate Request, Synthesize AI Prompt).
Solid Rectangle	Data Store	Repository where data is held for later querying (e.g., Supabase PostgreSQL, Audit Logs).
Labeled Arrow	Data Flow	Movement of data between entities, processes, and stores labeled with data payloads.

Level 1 System Data Flow Breakdown:

Process #	Process Name	Input Flow	Data Store / Entity	Output Flow
1.0	Auth & Session Management	Login credentials / JWT token	D1: User Profiles Store	Verified claims & user_id
2.0	Trip Validation & Prompt Engineering	Destination, dates, budget, vibe tags	Internal Schema Generator	Validated system & user prompt
3.0	Groq LPU Synthesis & Parsing	Engineered prompt payload	External: Groq Cloud LPU	Sub-second structured JSON itinerary
4.0	Relational Persistence	Parsed JSON trip object	D2: Trips & Itineraries DB D3: AI Audit Logs	Persisted records via SQLAlchemy
5.0	Timeline Customization & Re-roll	Updated slot or single-day re-roll	D2: Trips & Itineraries DB	Synchronized updated itinerary
6.0	Offline Export Engine	Print request & active trip state	Client Print DOM Formatter	Downloadable offline PDF guide